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Case 26-2025: An 11-Year-Old Girl with Chest Pain and Bone and Liver Lesions

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PRESENTATION OF CASE

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CME



Dr. Marcus L. Basiri (Pediatrics): An 11-year-old girl was transferred to the pediatric service of this hospital because of chest pain, arm pain, and multiple bone and liver lesions seen on imaging.

The patient had been in her usual state of health until 7 days before this presentation, when nausea, vomiting, and colicky abdominal pain developed, along with atraumatic pain in the right forearm, elbow, upper arm, and shoulder and in the sternal chest. Swelling of the right elbow was present; she had no fever or rigors. During the next few days, the nausea, vomiting, and abdominal pain abated, but the pain in the right arm and chest progressed, and she had discomfort when she took deep breaths. The patient's mother took her to the emergency department of another hospital for evaluation.

On presentation to the other hospital, the oral temperature was 37°C, the blood pressure 109/53 mm Hg, the heart rate 97 beats per minute, the respiratory rate 26 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. No axillary, inguinal, or cervical lymphadenopathy was noted on palpation. Cardiopulmonary auscultation was unremarkable. The abdomen was nontender, and there was no hepatosplenomegaly. The right arm and shoulder were tender on palpation but otherwise unremarkable, without visible joint effusions. No rash was present. Neurologic examination showed no motor or sensory deficits.

The blood levels of electrolytes, total bilirubin, aspartate aminotransferase, and alanine aminotransferase were normal, as were the results of kidney-function tests. An electrocardiogram showed sinus rhythm without ST-segment or T-wave abnormalities.

Dr. Sjirk J. Westra: Radiographs of the chest and the right forearm, elbow, and shoulder were normal (Fig. 1A). Venous ultrasonography of the right upper arm showed no evidence of deep-vein thrombosis but was notable for two prominent lymph nodes in the posterior elbow. There was no effusion in the right elbow joint or right axillary lymphadenopathy. Ultrasonography of the right upper abdomen showed a nonspecific, well-circumscribed, hypoechoic lesion (measuring 9 mm in

diameter) in the peripheral inferior right hepatic lobe (Fig. 1B).

Magnetic resonance imaging (MRI) of the right upper arm, elbow, and shoulder was performed with the administration of gadolinium-based contrast material. There was a lesion in the right humeral head that was hyperintense on T2-weighted imaging (Fig. 2A) and was associated with contrast enhancement (Fig. 2B). No subperiosteal or intraosseous collections were present. MRI of the chest revealed a T2-hyperintense bone marrow-replacing lesion in the sternum (Fig. 2C). Whole-body MRI revealed two small T2-hyperintense lesions in the right hepatic lobe (Fig. 2D).

Dr. Basiri: Treatment with naproxen was initi-

ated, and the patient's pain abated. Owing to the multifocal bone and liver lesions seen on imaging, she was transferred to this hospital for further evaluation.

On arrival at this hospital, the patient reported no further pain. She had normal vital signs with no fever; the remainder of the physical examination was normal. She did not report easy bruising or bleeding, night sweats, weight loss, anorexia, or rash. Her medical history was notable for eczema, as well as recurrent skin and soft-tissue infections due to methicillin-resistant *Staphylococcus aureus* (MRSA), for which she had most recently been treated with trimethoprim-sulfamethoxazole 2 months before this presentation. She took no medications and was up to date with all routine childhood immunizations. There were no known adverse reactions to medications. She lived in a suburb in Massachusetts with her mother and grandmother. She reported no recent travel or recent changes in dietary habits. She had not consumed raw, undercooked, or unpasteurized foods. The family had recently gotten a kitten; there were no other pets in the home.

Dr. Westra: Subsequent MRI of the upper abdomen confirmed the presence of two T2-hyperintense lesions in the liver (Fig. 3A) and revealed an additional T2-hyperintense lesion in the spleen (not shown). The hepatic lesion that had been seen on ultrasonography had peripheral (rim) enhancement during the portal venous phase (Fig. 3B and 3C). Diffusion-weighted imaging revealed numerous punctate hyperintense lesions throughout the liver (Fig. 3D, 3E, and 3F), which had not been detected on any of the routine sequences. The literature supports that diffusion-weighted imaging is more sensitive for the detection of liver lesions than conventional T2-weighted imaging, with or without the administration of contrast material.^{1,2} Only the largest of these lesions (Fig. 3F) had been visible on ultrasonography.

Dr. Basiri: Serologic testing was negative for human immunodeficiency virus (HIV) infection and *Toxoplasma gondii* infection. Other laboratory test results are shown in Table 1.

A diagnostic test was performed.

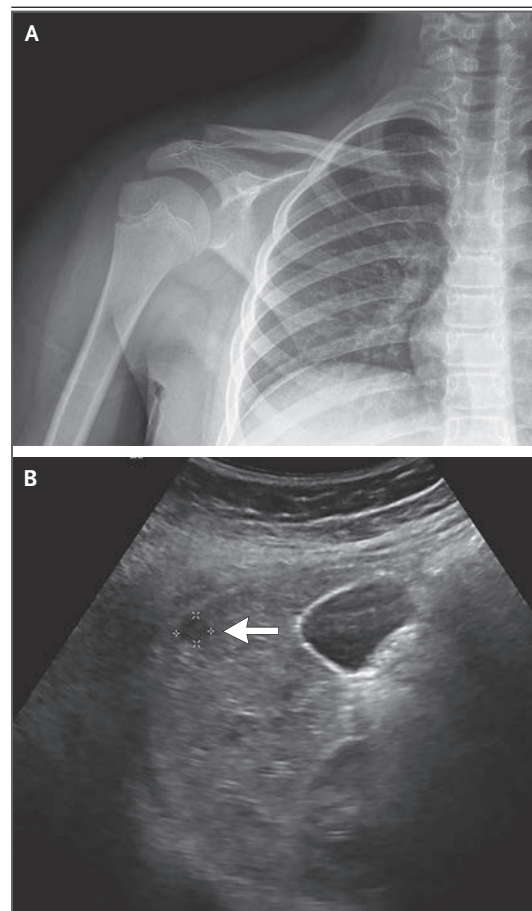


Figure 1. Initial Imaging Studies of the Right Shoulder and Liver.

A radiograph of the right shoulder is normal (Panel A). An ultrasound image of the liver shows a subcentimeter hypoechoic lesion in the peripheral right hepatic lobe (Panel B, arrow).

DIFFERENTIAL DIAGNOSIS

Dr. Lindsay P. Carter: This 11-year-old, well-appearing girl with a history of eczema and MRSA skin infec-

tions presented with pain in the right arm and sternum that had lasted for 7 days. The patient's examination was unremarkable except for tender-

ness in the right arm and shoulder. Laboratory test results were notable for elevated levels of inflammatory markers, microcytic anemia, and hypoa-

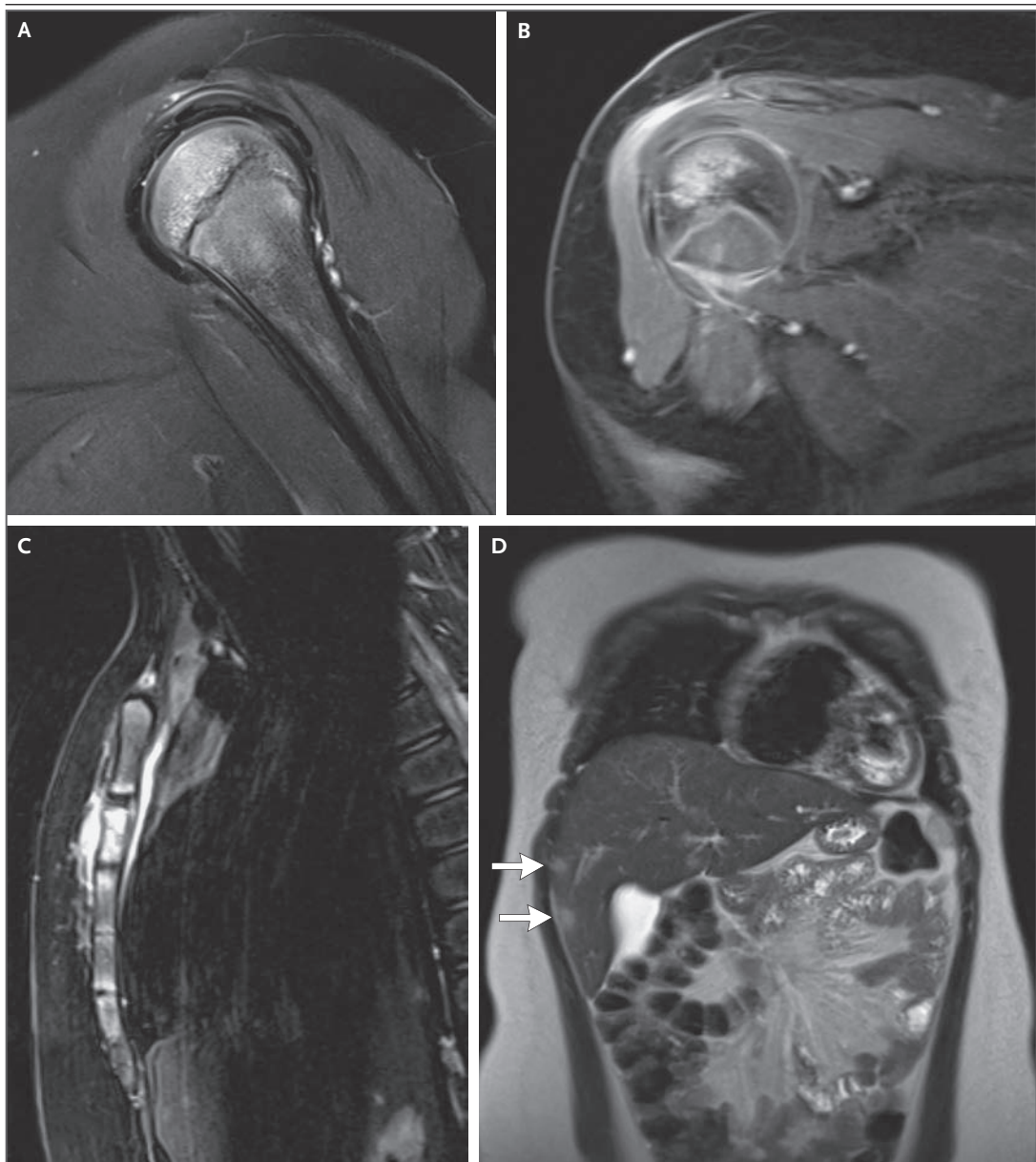


Figure 2. Initial MRI.

MRI of the right shoulder was performed. A fat-suppressed T2-weighted image (Panel A) shows a hyperintense lesion in the right humeral head. A fat-suppressed T1-weighted image, obtained after the administration of intravenous contrast material (Panel B), shows abnormal enhancement. MRI of the chest was also performed. A fat-suppressed T2-weighted image in a midsagittal plane (Panel C) shows an abnormal signal in the bone marrow, centered over the second sternal element, with surrounding soft-tissue involvement. Subsequent whole-body MRI was performed. A non-fat-suppressed T2-weighted image from half-Fourier single-shot turbo spin-echo imaging in a coronal plane (Panel D) shows two hyperintense lesions in the right hepatic lobe (arrows), with the most superior lesion corresponding to the lesion detected on ultrasonography. No other lesions were seen.

buminemia. Imaging revealed multifocal bone involvement without bone destruction, as well as epitrochlear lymphadenopathy and a rim-enhancing liver lesion. She had no fever, murmur, arthritis, joint effusion, systemic lymphadenopathy, organo-

megaly, or leukocytosis. On the basis of these findings, I will develop a differential diagnosis focusing on cancer, inflammatory disorders, and infection as the possible causes of this patient's bone and liver lesions and regional lymph-node involvement.

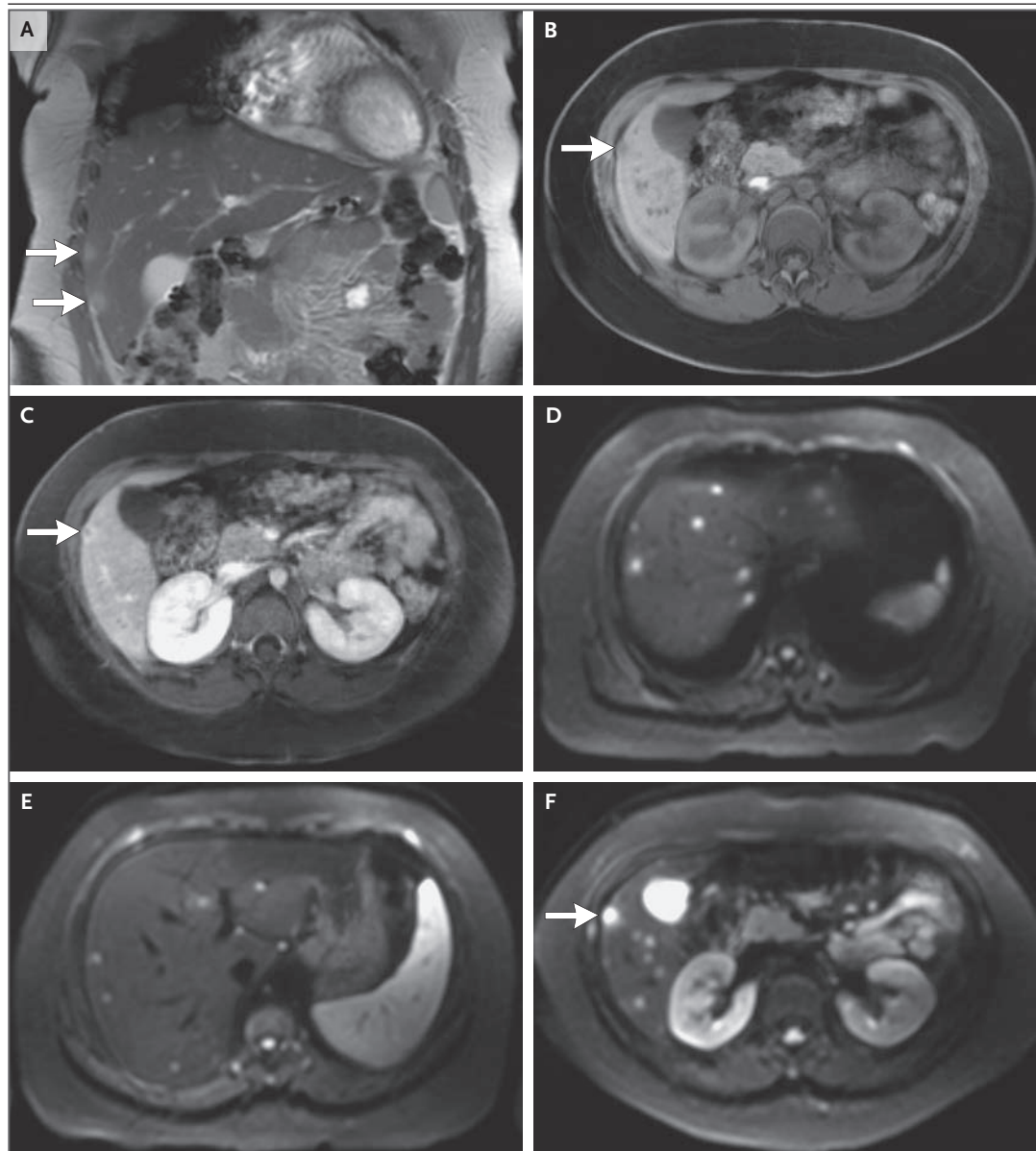


Figure 3. MRI of the Liver.

A coronal T2-weighted image from true fast imaging with steady-state precession (Panel A) confirms the presence of two hyperintense liver lesions (arrows). Axial fat-suppressed T1-weighted images of the most superior lesion, obtained before and after the administration of intravenous contrast material (Panels B and C, respectively), show peripheral (rim) enhancement of the hypointense lesion (arrows). Axial diffusion-weighted images of the liver (Panels D, E, and F) show numerous subcentimeter hyperintense lesions throughout the liver; the largest of these lesions, which corresponds to the sonographic abnormality, is identified with an arrow.

Table 1. Laboratory Data.*

Variable	Reference Range, Children†	On Presentation, This Hospital
Hemoglobin (g/dl)	11.2–14.5	10.3
Hematocrit (%)	34.3–43.0	32.0
Mean corpuscular volume (fl)	78.3–87.7	71.4
White-cell count (per μ l)	4000–10,000	9720
Differential count (per μ l)		
Neutrophils	1960–5690	3370
Lymphocytes	1790–3730	1960
Monocytes	350–780	410
Eosinophils	50–410	450
Basophils	20–60	30
Platelet count (per μ l)	205,000–376,000	467,000
Sodium (mmol/liter)	135–145	135
Potassium (mmol/liter)	3.4–5.0	3.8
Chloride (mmol/liter)	98–108	98
Carbon dioxide (mmol/liter)	23–32	23
Urea nitrogen (mg/dl)	9–23	11
Creatinine (mg/dl)	0.50–1.30	0.53
Calcium (mg/dl)	8.5–10.5	9.1
Aspartate aminotransferase (U/liter)	6–40	22
Alanine aminotransferase (U/liter)	10–49	17
Alkaline phosphatase (U/liter)	56–285	174
Total bilirubin (mg/dl)	0.0–1.2	0.2
Albumin (g/dl)	3.3–5.0	2.4
Globulin (g/dl)	1.9–4.1	4.3
Creatine kinase (U/liter)	33–211	106
High-sensitivity troponin T (ng/liter)	0–9	<6
Erythrocyte sedimentation rate (mm/hr)	45–115	>130
C-reactive protein (mg/liter)	0.0–8.0	40.1
Iron (μ g/dl)	50–170	50
Transferrin saturation (%)	13–45	15
Total iron-binding capacity (μ g/dl)	250–450	280
Ferritin (μ g/liter)	Variable	320

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for iron and iron-binding capacity to micromoles per liter, multiply by 0.1791.

† Reference values are affected by many variables, including the patient population and the laboratory methods used.

CANCER

This patient presented with disease involvement at multiple sites, including bone, liver, and possibly lymph nodes, prompting consideration of various

childhood cancers that could explain these findings. The most common bone tumors in children are Ewing's sarcoma and osteosarcoma. However, these tumors originate in the metaphyses of the

long bones, and this patient's lesions were located in the epiphyses. In addition, no evidence of soft-tissue masses, periosteal reactions, or cortical destruction was identified on imaging. Furthermore, these tumors often metastasize to the lungs, and this patient had normal results on pulmonary imaging, which also argues against a diagnosis of primary bone cancer.

Blood cancer such as leukemia should be considered in this case. Acute lymphoblastic leukemia is the most common cancer in children, but more than half of patients present with organomegaly, pallor, fever, or bruising³; none of these features were present in this patient. Epitrochlear lymphadenopathy, an enhanced signal in the bone marrow, and liver involvement can all occur with leukemia. However, if this patient had leukemia with liver involvement, I would expect the results of the complete blood count to be abnormal. In addition, whole-body imaging showed no systemic lymphadenopathy, splenomegaly, or mass that would suggest a diagnosis of lymphoma.

Langerhans'-cell histiocytosis, a multisystem histiocytic disorder that frequently involves the bones, skin, and liver, should also be considered. This patient had a history of a chronic inflammatory dermatosis that was thought to be eczema, but could it actually indicate Langerhans'-cell histiocytosis? Although this diagnosis is unlikely, it merits consideration to avoid cognitive biases such as anchoring and diagnostic momentum. However, the most common age at diagnosis is 1 to 3 years, the most common area of involvement is the skull, and lytic lesions or periosteal reactions on imaging would be expected.⁴

INFLAMMATORY DISORDERS

In a child presenting with multifocal bone involvement, chronic nonbacterial osteomyelitis should be considered. Chronic nonbacterial osteomyelitis is a noninfectious inflammatory disorder characterized by erosive bone inflammation that can lead to bone destruction at multiple sites if not treated. The diagnosis can be made with the use of either the Jansen criteria or the Bristol criteria.^{5,6} Some features of this patient's presentation fit these criteria: the bones involved (humerus and sternum), her well appearance, and the absence of fever. However, the high C-reactive protein level is not consistent with this diagnosis on the basis of the Bristol criteria, and she did not have coexisting

conditions or a family history that would meet the Jansen criteria. The most striking feature in this patient's case that does not fit with a diagnosis of chronic nonbacterial osteomyelitis is liver involvement. I considered the possibility of co-occurring extraintestinal inflammatory bowel disease, which has been reported in patients with chronic nonbacterial osteomyelitis,⁷ but the patient's normal growth makes these diagnoses unlikely.

Sarcoidosis is another entity that can involve bone and liver. However, osseous sarcoidosis occurs in less than 15% of patients with sarcoidosis and is most likely to involve the hands and feet.⁸ In addition, this patient was much younger than the typical patient with sarcoidosis, and she did not have evidence of bilateral hilar lymphadenopathy or parenchymal pulmonary disease on imaging, findings that are present in up to 90% of patients with sarcoidosis.⁹

The last entity considered in this category would require a shift in the framework of the differential diagnosis, but I think it is important to consider, given the patient's history and presentation. What if she had an underlying immunodeficiency such as chronic granulomatous disease? She had recurrent skin infections, osteomyelitis, liver abscess, elevated levels of inflammatory markers, and anemia of chronic disease, and she had received at least one course of trimethoprim-sulfamethoxazole. Chronic granulomatous disease is typically X-linked (in >50% of cases), but there are autosomal recessive types that may be diagnosed later in life and have less severe presentations. However, the patient did not have a history of pneumonia, which would be common in a patient with chronic granulomatous disease, and she had normal growth and did not have lymphadenopathy or organomegaly. Therefore, the diagnosis is unlikely.

INFECTION

In an otherwise healthy child, an acute illness such as that seen in this case is most often caused by infection. Bone involvement suggests the possibility of bacterial osteomyelitis. Her history of eczema and MRSA infections increases the probability of hematogenous spread of *S. aureus* to both bone and liver that resulted in hematogenous osteomyelitis and multifocal liver abscesses. However, children with acute hematogenous staphylo-

coccal osteomyelitis typically present with fevers, leukocytosis, and an ill appearance. Despite the absence of pulmonary involvement and epidemiologic risk factors, granulomatous infections due to tuberculous or nontuberculous mycobacteria or due to an endemic mycosis are worth considering in this case.

Extrapulmonary tuberculosis occurs in approximately one third of pediatric patients with tuberculosis but most commonly involves cervical lymph nodes. If this patient had tuberculosis involving several sites, I would expect her to have constitutional symptoms such as weight loss, night sweats, or fever.¹⁰ Although nontuberculous mycobacteria are ubiquitous in the environment and infections often develop in immunocompetent children, such infections are typically seen in children younger than this patient and usually involve the cervicofacial region.¹¹ Systemic nontuberculous mycobacterial infection is mostly seen in immunocompromised patients, and children presenting with this diagnosis are often acutely ill.

Endemic mycoses such as histoplasma and blastomyces infections typically first involve the lungs, given that the spores are inhaled, and can disseminate in immunocompromised hosts to distant sites, including bone and liver. Although histoplasmosis occurs sporadically in New England, both histoplasmosis and blastomycosis are mostly geographically restricted. The patient had no known history of travel to areas where the infections are endemic, such as the Midwest or the southeastern United States. Finally, this patient had negative tests for HIV infection and toxoplasmosis, and therefore, these entities can be removed from the differential diagnosis.

Given the features of this patient's presentation and recent exposure to a kitten, infection with *Bartonella henselae* (cat scratch disease) is the most likely diagnosis in this case. More than 90% of kittens younger than 1 year of age have antibody reactivity to *B. henselae*.¹² The typical portal of entry for bartonella is inoculation through a kitten scratch, kitten bite, or flea bite. After infection, regional lymphadenopathy develops at the site of inoculation within 1 to 3 weeks. Cat scratch disease is usually a self-limited illness, and up to 50% of patients are afebrile.^{13,14} However, in patients with systemic disease, bartonella can spread to the visceral organs, including the liver, bones, eyes, and neurologic system, and the infection is often

accompanied by fever. In fact, cat scratch disease is the third most common cause of fever among children presenting with fever of unknown origin.¹⁵ *B. henselae* preferentially infects the bones of the lower part of the body and the spine; however, the infection can affect any long bone, including the humerus, and can involve the sternum. Cat scratch disease can cause hepatosplenic disease with microabscesses, which would be consistent with the liver lesions seen on imaging in this patient.

Given the patient's well appearance and a desire not to put her through unnecessary invasive testing such as bone biopsy, I would perform serologic testing for bartonella infection to obtain a diagnosis. I would begin empirical antimicrobial therapy while waiting for the test results.

CLINICAL IMPRESSION

Dr. Ann M. Murray: This patient's key epidemiologic risk factors for infection included exposure to a kitten and a history of recurrent MRSA soft-tissue infections and eczema. Given the previous antibiotic treatments for skin infections, an atypical presentation of partially treated acute hematogenous osteomyelitis and liver abscesses from bacteremia was considered, and blood cultures were obtained. However, she had no fever and the pain resolved without antibiotic therapy, factors that made infection with *S. aureus* or another pathogen typically associated with osteomyelitis unlikely. Brucellosis, salmonella infection, HIV infection, and tuberculosis were also considered, but there were no identified risk factors for these infections, such as exposure to persons with tuberculosis or HIV infection, a relevant travel history, exposure to reptiles, or consumption of raw, undercooked, or unpasteurized foods. The multiple osseous abnormalities seen on imaging in the right humerus and sternum were not associated with subperiosteal or intraosseous collections and were located in atypical sites for pediatric osteomyelitis. Approximately 70% of cases of osteomyelitis in children involve one bone of the lower legs or hips.¹⁶

Bone lesions that occur with *B. henselae* infection are often multifocal and involve unusual sites, such as the spine, pelvic girdle, cranium, and sternum.¹⁷ In this patient, disseminated *B. henselae* infection was thought to be the most likely diagnosis, given the multiple bone lesions at atypical

sites combined with the presence of liver lesions, elevated levels of inflammatory markers, and exposure to a kitten at home. To provide diagnostic clarity, the initial plan was to perform a biopsy for pathological and microbiologic evaluation. However, the results of serologic testing for bartonella became available before the biopsy was performed.

CLINICAL DIAGNOSIS

Disseminated *Bartonella henselae* infection.

DR. LINDSAY P. CARTER'S DIAGNOSIS

Bartonellosis.

PATHOLOGICAL DISCUSSION

Dr. Sarah E. Turbett: Immunofluorescence antibody assays for *B. henselae* and *B. quintana* were performed at a reference laboratory.¹⁸ *B. henselae* IgM and IgG isotypes were detected at a titer of greater than 1:20 (reference value, negative) and 1:4096 (reference value, negative), respectively; *B. quintana* IgM and IgG isotypes were not detected.

B. henselae is a fastidious gram-negative bacterium that infects red cells and endothelial cells in humans. Its primary vector is *Ctenocephalides felis* (cat flea), and cats serve as the predominant reservoir.¹⁹ Humans are incidental hosts, with the infection occurring as a result of a bite or scratch from an infected cat; human-to-human transmission of bartonella is not typical.¹⁹

The most common manifestation of *B. henselae* infection is cat scratch disease, which is characterized by the development of a lesion at the site of inoculation, followed by regional lymphadenopathy and systemic symptoms.²⁰ Atypical manifestations occur in 5 to 20% of patients and can include ocular, neurologic, hepatosplenic, or musculoskeletal involvement.²⁰

Serologic testing remains the cornerstone of the diagnosis of cat scratch disease; in this case, the IgM titer of greater than 1:20 and the IgG titer of greater than 1:180 were suggestive of active bartonella infection.¹⁸ In general, an increase in serial IgG titers by a factor of 4 is also supportive of the diagnosis.¹⁹ Direct detection of the pathogen with molecular methods such as polymerase-chain-reaction (PCR) testing is also available in

most reference laboratories across the United States, with reports of high diagnostic sensitivity and specificity.²¹ For cat scratch disease, the sensitivity of PCR testing for bartonella is increased when the test is performed directly on tissue from the lesion of the primary inoculation site or on purulent material from an infected lymph node.²¹ Bartonella-specific tissue culture is of limited use owing to the fastidious nature of the bacteria; the median time to growth from an infected lymph node is 21 days.¹⁹

On the basis of the clinical presentation and serologic test results in this case, the microbiologic diagnosis is *B. henselae* infection.

MICROBIOLOGIC DIAGNOSIS

Bartonella henselae infection.

DISCUSSION OF MANAGEMENT

Dr. Murray: The management of typical cat scratch disease resulting from *B. henselae* infection primarily involves supportive care, and the disease is usually self-limited, resolving spontaneously in 2 to 4 months. A 5-day course of azithromycin for bartonella lymphadenitis was evaluated in a prospective, double-blind, randomized, placebo-controlled investigation in 1998 and had significant clinical benefit.²² However, this studied treatment for lymphadenitis would not address this patient's atypical systemic presentation. The treatment of systemic *B. henselae* infection is guided by case studies owing to a lack of randomized, controlled trials for this rare manifestation. A previous study indicated that rifampin, with 87% efficacy, was the most effective treatment.²³ The combination of rifampin and doxycycline has been reported to be effective for the treatment of *B. henselae*-induced neuroretinitis and other cases of bartonellosis with central nervous system involvement.²⁴ Case reports of bartonella osteomyelitis have also described successful treatment with the combination of rifampin and doxycycline.^{25,26}

In this case, an ophthalmology consultation was requested, and the patient was found to have no evidence of intraocular infection. She also had no central nervous system involvement. A 28-day course of oral doxycycline and rifampin was prescribed. She was discharged home with a plan for follow-up in the outpatient infectious diseases

clinic, and she was advised to use oral naproxen as needed for pain.

FOLLOW-UP

Dr. Murray: The patient completed the 28-day course of doxycycline and rifampin, and she reported no missed doses or side effects. At her follow-up appointment, she reported no bone pain, abdominal pain, fevers, vision changes, or headaches. There were no new symptoms. She had started treatment with dupilumab, which had been prescribed by her dermatologist for eczema control, and her skin symptoms had abated. Follow-up laboratory studies performed after completion of the antibiotic course were notable for normalization of the C-reactive protein level and improvement of the erythrocyte sedimentation rate (from >130 to 68 mm per hour).

Serologic testing for bartonella performed during the convalescent phase of illness showed a persistently elevated *B. henselae* IgG titer (1:4096), but the *B. henselae* IgM titer (<1:20) was within the normal range; these findings are consistent with seroconversion. The alanine aminotransferase and aspartate aminotransferase levels remained normal, and mild anemia persisted. Results of an initial immune screening — with measurement of lymphocyte subsets, quantitative immunoglobulins, and vaccine titers — were reassuringly nor-

mal, other than low IgG titers related to the pneumococcal vaccine.

Repeat whole-body MRI unexpectedly showed that hepatic lesions and two splenic lesions had increased in size, and new lesions were detected in the bilateral femora, tibiae, carpal bones, and right humeral epiphysis. The patient started combination therapy with azithromycin and rifampin, with a plan for a 6-month course and a formal immunologic evaluation. She remained asymptomatic, and she was reassured that the prognosis for all forms of cat scratch disease is excellent. Even without antibiotic treatment, almost all immunocompetent patients fully recover, and recurrence of disease is rare.¹⁷

FINAL DIAGNOSIS

Disseminated bartonellosis.

This case was presented at Pediatric Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

AUTHOR INFORMATION

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